United International University

Department of Computer Science & Engineering

Final Exam, Trimester: Summer 2026

Course Code: CSE 3715 **Course Title:** Data Communication

Duration: 2 hours **Total Marks:** 40

(Any examinee found adopting unfair means would be expelled from the trimester/program as per UIU disciplinary rules.)

ANSWER ALL THE QUESTIONS

1. A telemedicine company runs two systems- system A sends live video calls using circuit switching with synchronous TDM, and system B sends patient records using packet switching.

   (a) In system A, four video channels, each operating at 2 Mbps, are multiplexed using synchronous TDM. Each time slot carries 2 bits from each channel. Assume there are no framing bits. **Find:** **[4]**

   (i) the duration of one input bit,

   (ii) the number of bits in each frame,

   (iii) the frame rate,

   (iv) the data rate of the multiplexed link.

   (b) Suppose some channels in system A are often idle. **Identify** what problem this causes in synchronous TDM. **State** two differences between synchronous TDM and statistical TDM. **[3]**

   (c) In system B, a patient record of $50{,}000$ bits is sent over a path with 3 routers (so, 4 links). Each router has a processing time of 0.1 ms. Each link is 800 km long, has a bandwidth of 10 Mbps, and the signal speed is 2×10^8 m/s. **Calculate** the total end-to-end delay. **[3]**

2. The network system adds redundant bits so the receiver can detect, and sometimes correct, errors and protect patient data,

   (a) **Illustrate** how redundancy is used to detect errors in error control mechanisms. **[2]**

   (b) **Compute** the 2D even-parity bits for the 4×4 block: 1011, 0110, 1110, 0001. **Show** one error pattern that 2D parity **cannot** detect. **[3]**

   (c) The system is then tested using CRC with the generator polynomial $G(x) = x^5 + x^3 + x + 1$. The received data is: **[4]**

   $$x^{13} + x^{11} + x^9 + x^6 + x^4 + x^3 + x$$

   Perform error detection at the receiver and **decide** whether the frame will be accepted or rejected.

   (d) The network administrators now encode 4-bit data blocks into 7-bit codewords using Hamming code C(7,4). One received codeword is: **[3]**

   $$0110100$$

   Detect and **correct** the error, if any, and **write** the original 4-bit data.

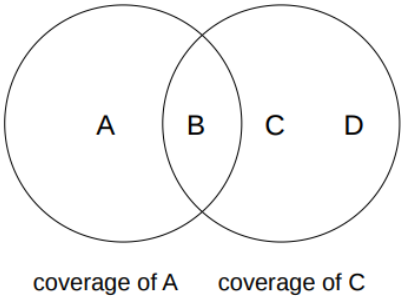
(e) **Explain** how the number of parity bits is determined in a Hamming code. **Compute** the parity bits needed for 15 data bits and **list** their bit positions. **[3]**

3. On campus, the computer lab uses wired Ethernet, whose access method **detects** collisions. The library uses Wi-Fi, whose access method **avoids** collisions. Both methods grew out of ALOHA, an early random-access protocol for radio and satellite links, which was later improved as slotted ALOHA.

(a) **Identify** the access method used in the library. **Explain** how it avoids collisions. **[3]**

(b) **Explain** why the access method used in the computer lab cannot be used in the library. **[2]**

(c) In the library's Wi-Fi network, station A is sending data to station B. At the same time, station C wants to send data to B. A and C are out of each other's range, as shown below. **[3]**

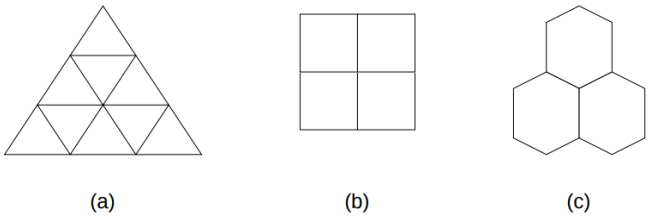


Determine whether a collision will occur. **Explain** how the library's access method handles this situation, and **show** it using a timing diagram.

(d) **Describe** how slotted ALOHA improves the efficiency of pure ALOHA. **[2]**

4. In cellular network system, a geographical area is divided into many small cells. A group of cells that together use all available frequencies is called a cluster.

(a) **Select** the shape that best represents a cell and **justify** your choice. **[1]**



(b) **Explain** the reason for dividing a geographical area into many small cells rather than using a single large cell. **[2]**

(c) A operator uses a cluster size of **N = 7**, and each cell has a radius of **2 km**. **Calculate** the frequency reuse distance between two cells that use the same set of frequencies. **[2]**